

COMMON PROJECTION CENTERS FOR $d + 3$ POINT PAIRS: A KERNEL PARAMETRIZATION OF THE FRONTIER CASE

ABSTRACT. Let $X = (x_1, \dots, x_N)$ and $Y = (y_1, \dots, y_N)$ be general ordered configurations in two copies of $\mathbf{P}^d$. We study pairs of centers $(a, b) \in \mathbf{P}^d \times \mathbf{P}^d$ for which projection from a and projection from b give projectively equivalent ordered configurations in $\mathbf{P}^{d-1}$.

First, we construct the universal incidence variety, prove that its relevant open part is irreducible, and deduce generic emptiness whenever

$$D(d, N) = d^2 + 2d - 1 - (d - 1)N < 0.$$

We then solve the uniform frontier case $N = d + 3$. If X and Y are represented by full-rank configuration matrices, their kernels K_X, K_Y are two-dimensional. A pair of lines $([k], [\ell]) \in \mathbf{P}(K_X) \times \mathbf{P}(K_Y)$ determines an invertible diagonal matrix $\text{diag}(k_i/\ell_i)$, and hence a pair of camera centers. This gives a birational parametrization of the centers-variety by $\mathbf{P}^1 \times \mathbf{P}^1$.

The parametrization has exactly N natural base points. Blowing them up resolves the map and gives the pullback hyperplane classes

$$H_a = H_1 + (N - 2)H_2 - \sum_{i=1}^N E_i, \quad H_b = (N - 2)H_1 + H_2 - \sum_{i=1}^N E_i.$$

Consequently, for $N = d + 3$, the centers-variety is an irreducible rational surface of multidegree

$$(d - 1, d^2 + d - 1, d - 1).$$

We then settle the principal questions left open in the first version. An elementary-modification calculation shows that the two one-center surfaces are the balanced rational normal scrolls

$$S\left(\left\lfloor \frac{d-1}{2} \right\rfloor, \left\lceil \frac{d-1}{2} \right\rceil\right).$$

For $d \geq 3$, each individual projection contracts exactly $d + 3$ explicit disjoint (-1) -curves, while the pair map separates all points and tangent vectors. It is therefore a closed immersion. In particular, the centers-variety is smooth and isomorphic to the blow-up of either balanced scroll at $d + 3$ points.

In the first new case $d = 4, N = 7$, its Chow class is

$$3h_a^4 h_b^2 + 19h_a^3 h_b^3 + 3h_a^2 h_b^4,$$

its degree in the Segre embedding is 44, and the one-center loci are smooth cubic scrolls $S(1, 2) \subset \mathbf{P}^4$. We also give a prime elimination presentation of the bihomogeneous ideal and the graph linear system on the cubic scroll. Determining minimal ideal generators and the full bigraded Betti table remains computational work.

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1. RESEARCH QUESTION AND CURRENT STATUS

1.1. **The problem.** Throughout the main arguments, $\mathbf{k}$ is an algebraically closed field of characteristic zero. Let V and U be vector spaces of dimension $d + 1$, and let

$$X = (x_1, \dots, x_N) \in \mathbf{P}(V)^N, \quad Y = (y_1, \dots, y_N) \in \mathbf{P}(U)^N.$$

For $a \in \mathbf{P}(V)$, let $\widehat{a} \subset V$ be the corresponding one-dimensional vector subspace. Projection from a is the quotient map

$$q_a : \mathbf{P}(V) \dashrightarrow \mathbf{P}(V/\widehat{a}) \cong \mathbf{P}^{d-1}.$$

Define q_b similarly.

Definition 1.1 (Open centers locus and centers-variety). The open centers locus is

$$\begin{aligned} \mathcal{C}_{d,N}^\circ(X, Y) := \{ (a, b) \in \mathbf{P}(V) \times \mathbf{P}(U) : a \notin \{x_1, \dots, x_N\}, b \notin \{y_1, \dots, y_N\}, \\ \exists \phi : \mathbf{P}(V/\widehat{a}) \xrightarrow{\sim} \mathbf{P}(U/\widehat{b}), \\ \phi(q_a(x_i)) = q_b(y_i) \quad (1 \leq i \leq N) \}. \end{aligned}$$

Its Zariski closure

$$\mathcal{C}_{d,N}(X, Y) := \overline{\mathcal{C}_{d,N}^\circ(X, Y)} \subset \mathbf{P}(V) \times \mathbf{P}(U)$$

is the centers-variety.

The research plan selected the first new case

$$d = 4, \quad N = 7,$$

and the uniform frontier $N = d + 3$. The primary question was whether $\mathcal{C}_{d,d+3}(X, Y)$ is generically nonempty and pure of dimension 2, and, if so, to determine its components and its class in

$$A^*(\mathbf{P}^d \times \mathbf{P}^d) = \mathbb{Z}[h_a, h_b]/(h_a^{d+1}, h_b^{d+1}).$$

1.2. What is proved in this note.

Statement	Status	Main input
Universal incidence is irreducible and has the predicted dimension	Proved	Relative projective-isomorphism bundle and affine-line fibers
Generic emptiness when $D(d, N) < 0$	Proved	Dimension of the universal incidence image
$\mathcal{C}_{d,N} = \mathbf{P}^d \times \mathbf{P}^d$ for $N \leq d + 1$	Proved	Transitivity on projective frames; closure is essential
$N = d + 3$: nonempty, irreducible, rational surface	Proved	Two-dimensional kernels and a diagonal-kernel criterion
Resolution by blowing up $d + 3$ points of $\mathbf{P}^1 \times \mathbf{P}^1$	Proved	Explicit bihomogeneous formulas and local base-point analysis
Multidegree $(d - 1, d^2 + d - 1, d - 1)$	Proved	Intersection theory on the blow-up
Projection loci are degree $d - 1$ rational normal scrolls	Proved	Minimal-degree theorem and the line ruling
Balanced scroll type for every d	Proved	Maximal-rank interpolation for an elementary modification
Closed immersion, smoothness, and normality for $d \geq 3$	Proved	Classification of contracted curves and tangent separation
Minimal ideal generators and bi-graded Betti table	Open	Prime elimination presentation is proved; minimal computation remains

1.3. **Corrections to the starting draft.** The earlier three-dimensional draft contained the correct expected dimension but treated several consequences of the count as if they were unconditional. The following distinctions are needed.

- (i) A parameter count does not prove dominance, nonemptiness, purity, or transversality.
- (ii) When $N \leq d + 1$, the *closure* of the valid locus is $\mathbf{P}^d \times \mathbf{P}^d$. Centers equal to data points are not themselves valid cameras.
- (iii) Bilinear or Grassmann-tensor constraints are necessary, but a compatibility theorem is required before an arbitrary tensor solution may be called a camera pair.
- (iv) The kernel criterion below is exact and avoids the compatibility gap.

2. THE UNIVERSAL INCIDENCE VARIETY

2.1. **Construction.** Let $\mathcal{B}_d$ parametrize triples (a, b, ϕ) , where

$$a \in \mathbf{P}(V), \quad b \in \mathbf{P}(U), \quad \phi : \mathbf{P}(V/\widehat{a}) \xrightarrow{\sim} \mathbf{P}(U/\widehat{b}).$$

It is a relative $\mathrm{PGL}(d)$ -bundle over $\mathbf{P}(V) \times \mathbf{P}(U)$. Hence

$$\dim \mathcal{B}_d = d + d + (d^2 - 1) = d^2 + 2d - 1.$$

For a fixed (a, b, ϕ) , define the matched-pair space

$$\mathcal{R}_{a,b,\phi} := \{(x, y) \in (\mathbf{P}(V) \setminus \{a\}) \times (\mathbf{P}(U) \setminus \{b\}) : \phi(q_a(x)) = q_b(y)\}.$$

Lemma 2.1. $\mathcal{R}_{a,b,\phi}$ is smooth and irreducible of dimension $d + 1$.

Proof. Projection from a ,

$$\mathbf{P}(V) \setminus \{a\} \longrightarrow \mathbf{P}(V/\widehat{a}),$$

is an affine-line bundle: over a point of $\mathbf{P}(V/\widehat{a})$, its fiber is a projective line through a with the point a removed. The same holds for projection from b . After identifying the two bases using ϕ , $\mathcal{R}_{a,b,\phi}$ is the fiber product of two affine-line bundles over $\mathbf{P}^{d-1}$. It is therefore an affine-plane bundle over $\mathbf{P}^{d-1}$. Thus it is smooth and irreducible, and

$$\dim \mathcal{R}_{a,b,\phi} = (d-1) + 2 = d+1.$$

□

Definition 2.2 (Universal incidence). Let $\mathcal{I}_{d,N}^\circ$ be the space of tuples

$$(X, Y, a, b, \phi)$$

such that $x_i \neq a$, $y_i \neq b$, and $\phi(q_a(x_i)) = q_b(y_i)$ for all i .

Theorem 2.3 (Universal incidence theorem). *The variety $\mathcal{I}_{d,N}^\circ$ is smooth and irreducible, and*

$$\dim \mathcal{I}_{d,N}^\circ = d^2 + 2d - 1 + N(d+1).$$

Consequently, if

$$D(d, N) := d^2 + 2d - 1 - (d-1)N < 0,$$

then a general pair

$$(X, Y) \in \mathbf{P}(V)^N \times \mathbf{P}(U)^N$$

has empty open centers locus.

Proof. The base $\mathcal{B}_d$ is irreducible. Over it, the choice of each matched pair (x_i, y_i) is a smooth irreducible family of relative dimension $d+1$ by Lemma 2.1. Taking the N -fold relative fiber product proves smoothness, irreducibility, and the dimension formula.

The data space $\mathbf{P}(V)^N \times \mathbf{P}(U)^N$ has dimension $2dN$. The difference between the dimension of the universal incidence and the dimension of the data space is

$$\begin{aligned} \dim \mathcal{I}_{d,N}^\circ - 2dN &= d^2 + 2d - 1 + N(d+1) - 2dN \\ &= d^2 + 2d - 1 - (d-1)N \\ &= D(d, N). \end{aligned}$$

If $D(d, N) < 0$, the image of $\mathcal{I}_{d,N}^\circ$ in the data space has strictly smaller dimension. Its closure is therefore a proper subvariety. Outside that subvariety, no matching cameras exist. □

Remark 2.4. When $D(d, N) \geq 0$, Theorem 2.3 gives no nonemptiness statement. One still has to prove that the projection to the data space is dominant. The kernel construction in Section 4 proves this dominance for $N = d+3$.

2.2. The automatic range.

Proposition 2.5. *Let X, Y be general. If $N \leq d+1$, then*

$$\mathcal{C}_{d,N}(X, Y) = \mathbf{P}(V) \times \mathbf{P}(U).$$

Proof. There is a dense open subset of $\mathbf{P}(V)$ consisting of centers a for which $q_a(x_1), \dots, q_a(x_N)$ are in linear general position in $\mathbf{P}^{d-1}$. Indeed, failure of linear general position places a in one of finitely many linear spans determined by subsets of X . The analogous statement holds for Y .

Any two ordered N -tuples in linear general position in $\mathbf{P}^{d-1}$ are projectively equivalent when $N \leq d+1$: extend each tuple to a projective frame and map one frame to the other. Thus a dense open subset of $\mathbf{P}(V) \times \mathbf{P}(U)$ lies in $\mathcal{C}_{d,N}^\circ(X, Y)$. Taking the Zariski closure gives the result. □

3. THE EXACT KERNEL CRITERION

Fix $N \geq d + 1$. Choose nonzero vector representatives of the data points and form the full-rank configuration maps

$$X : \mathbf{k}^N \longrightarrow V, \quad Y : \mathbf{k}^N \longrightarrow U,$$

whose i -th columns represent x_i and y_i . Set

$$K_X := \ker X, \quad K_Y := \ker Y.$$

For $a \in \mathbf{P}(V)$, define

$$L_X(a) := X^{-1}(\widehat{a}) \subset \mathbf{k}^N.$$

Equivalently,

$$L_X(a) = \ker(q_a X).$$

Define $L_Y(b)$ similarly.

Lemma 3.1 (Diagonal-kernel criterion). *Assume $a \notin X$ and $b \notin Y$. The projected ordered configurations $(q_a(x_i))_{i=1}^N$ and $(q_b(y_i))_{i=1}^N$ are projectively equivalent if and only if there is an invertible diagonal matrix*

$$D = \text{diag}(d_1, \dots, d_N)$$

such that

$$L_X(a) = D L_Y(b).$$

Proof. Suppose the projected configurations are equivalent. Then there are an isomorphism $H : V/\widehat{a} \rightarrow U/\widehat{b}$ and nonzero column scalars d_i such that

$$q_b Y = H(q_a X) D.$$

Taking kernels gives

$$L_Y(b) = D^{-1} L_X(a),$$

which is equivalent to the asserted equality.

Conversely, assume $L_X(a) = D L_Y(b)$. The two surjective maps

$$q_b Y \quad \text{and} \quad (q_a X) D$$

from $\mathbf{k}^N$ to d -dimensional vector spaces have the same kernel. They therefore induce two isomorphisms from the common quotient $\mathbf{k}^N / L_Y(b)$. Composing one with the inverse of the other gives an isomorphism H satisfying

$$q_b Y = H(q_a X) D.$$

Columnwise, this is exactly projective equivalence of the two ordered image configurations. $\square$

4. THE FRONTIER $N = d + 3$: KERNEL-LINE PARAMETRIZATION

From now on,

$$N = d + 3.$$

Since X and Y have rank $d + 1$,

$$\dim K_X = \dim K_Y = 2.$$

Thus

$$\mathbf{P}(K_X) \times \mathbf{P}(K_Y) \cong \mathbf{P}^1 \times \mathbf{P}^1.$$

4.1. **Genericity conditions.** We impose the following open conditions.

- (G1) The restriction of every coordinate functional on $\mathbf{k}^N$ to K_X and to K_Y is nonzero.
- (G2) On each of the two projective lines $\mathbf{P}(K_X)$ and $\mathbf{P}(K_Y)$, the zeros of the N coordinate functionals are pairwise distinct.
- (G3) There is no invertible diagonal matrix D such that $K_X = DK_Y$.

These conditions hold for a general pair of configurations. For (G3), note that the diagonal torus modulo scalars has dimension $N - 1$, whereas

$$\dim \operatorname{Gr}(2, N) = 2N - 4 > N - 1$$

for $N \geq 4$. Hence a general pair of two-planes does not lie in one diagonal-torus orbit.

4.2. **The construction.** Choose

$$[k] \in \mathbf{P}(K_X), \quad [\ell] \in \mathbf{P}(K_Y)$$

such that every coordinate k_i, ℓ_i is nonzero, and put

$$D(k, \ell) := \operatorname{diag} \left(\frac{k_1}{\ell_1}, \dots, \frac{k_N}{\ell_N} \right).$$

Choose $k' \in K_X \setminus \mathbf{k}k$ and $\ell' \in K_Y \setminus \mathbf{k}\ell$. Define

$$(1) \quad a(k, \ell) := [XD(k, \ell)\ell'] \in \mathbf{P}(V), \quad b(k, \ell) := [YD(k, \ell)^{-1}k'] \in \mathbf{P}(U).$$

Lemma 4.1. *The projective points in (1) are independent of the choices of k' and ℓ' .*

Proof. Replacing ℓ' by $\alpha\ell + \beta\ell'$, with $\beta \neq 0$, changes the first vector in (1) to

$$\alpha XD\ell + \beta XD\ell' = \alpha Xk + \beta XD\ell' = \beta XD\ell',$$

because $k \in K_X$. Thus its projective class is unchanged. The proof for $b(k, \ell)$ is symmetric. $\square$

Theorem 4.2 (Kernel-line parametrization). *Let X, Y satisfy (G1)–(G3). Formula (1) defines a rational map*

$$\Phi_{X,Y} : \mathbf{P}(K_X) \times \mathbf{P}(K_Y) \dashrightarrow \mathcal{C}_{d,d+3}(X, Y).$$

It maps a dense open subset into the open centers locus and is birational onto the centers-variety. In particular, $\mathcal{C}_{d,d+3}(X, Y)$ is nonempty, irreducible, rational, and of pure dimension 2.

Proof. Let $D = D(k, \ell)$. Since $D\ell = k$, we have

$$K_X \cap DK_Y \supseteq \mathbf{k}k.$$

Condition (G3) implies $K_X \neq DK_Y$, so

$$\dim(K_X + DK_Y) = 3.$$

By definition of $a = a(k, \ell)$,

$$\begin{aligned} L_X(a) &= X^{-1}(\mathbf{k}XD\ell') \\ &= K_X + \mathbf{k}D\ell' \\ &= K_X + DK_Y. \end{aligned}$$

Similarly,

$$L_Y(b) = K_Y + D^{-1}K_X.$$

Therefore,

$$DL_Y(b) = DK_Y + K_X = L_X(a).$$

Lemma 3.1 now proves that (a, b) is a valid pair of centers whenever the images are defined. The excluded conditions are closed, so a dense open subset of the parameter space maps into $\mathcal{C}_{d,d+3}^\circ(X, Y)$.

We prove birationality. Let (a, b) be a general valid center pair. By Lemma 3.1, there is an invertible diagonal D such that

$$L_X(a) = DL_Y(b).$$

This D is unique up to a scalar. Indeed, if D' is another choice, then $D^{-1}D'$ stabilizes $L_Y(b)$. If M is a full-rank matrix with kernel $L_Y(b)$, a diagonal stabilizer gives

$$M(D^{-1}D') = HM$$

for some $H \in \mathrm{GL}(d)$. Thus every column of M is an eigenvector of H . A general projected configuration contains a projective frame, so a projectivity fixing all its ordered points is the identity. Hence H is scalar, and so is $D^{-1}D'$.

Both two-planes

$$K_X \quad \text{and} \quad DK_Y$$

lie in the three-dimensional space $L_X(a)$. Their intersection is therefore nonzero. It is one-dimensional by (G3). Let

$$\mathbf{k}k = K_X \cap DK_Y, \quad k = D\ell.$$

Then $[k] \in \mathbf{P}(K_X)$ and $[\ell] \in \mathbf{P}(K_Y)$ are uniquely determined. The preceding construction recovers (a, b) . This gives a rational inverse on dense open subsets, proving birationality. $\square$

Corollary 4.3 (Dominance at the frontier). *For $N = d + 3$, the projection*

$$\mathcal{I}_{d,d+3}^\circ \longrightarrow \mathbf{P}(V)^{d+3} \times \mathbf{P}(U)^{d+3}$$

is dominant. Its general fiber has dimension

$$D(d, d + 3) = 2.$$

Proof. Theorem 4.2 constructs a two-dimensional family for a general pair (X, Y) , so the universal incidence dominates the data space. The dimension of a general fiber is the difference of the dimensions, which is $D(d, d + 3) = 2$. $\square$

Corollary 4.4 (Real centers). *Suppose X and Y are real and satisfy the same genericity conditions over $\mathbb{R}$. Then there is a two-real-parameter family of real valid center pairs.*

Proof. Both $\mathbf{P}(K_X)(\mathbb{R})$ and $\mathbf{P}(K_Y)(\mathbb{R})$ are real projective lines. Choose $[k]$ and $[\ell]$ outside the finite union of coordinate zeros and the remaining proper degeneracy loci. Formula (1) is then real. $\square$

5. EXPLICIT FORMULAS AND RESOLUTION OF THE PARAMETRIZATION

Choose bases

$$K_X = \mathrm{Span}(r, s), \quad K_Y = \mathrm{Span}(p, q),$$

and write

$$k(u, v) = ur + vs, \quad \ell(z, w) = zp + wq.$$

Thus k_i is linear in (u, v) and ℓ_i is linear in (z, w) .

Define V -valued bihomogeneous forms

$$\begin{aligned} \mathcal{A}_p &:= \sum_{i=1}^N x_i k_i p_i \prod_{j \neq i} \ell_j, \\ \mathcal{A}_q &:= \sum_{i=1}^N x_i k_i q_i \prod_{j \neq i} \ell_j. \end{aligned}$$

Since $Xk = 0$,

$$\begin{aligned} z\mathcal{A}_p + w\mathcal{A}_q &= \sum_{i=1}^N x_i k_i (zp_i + wq_i) \prod_{j \neq i} \ell_j \\ &= \left(\prod_{j=1}^N \ell_j \right) \sum_{i=1}^N x_i k_i \\ &= 0. \end{aligned}$$

Because z and w are coprime,

$$z \mid \mathcal{A}_q, \quad w \mid \mathcal{A}_p.$$

Hence

$$(2) \quad \mathbf{A} := \frac{\mathcal{A}_q}{z} = -\frac{\mathcal{A}_p}{w}$$

is a V -valued polynomial of bidegree

$$(1, N-2).$$

On the chart $z \neq 0$, q is a valid choice of ℓ' , and

$$[\mathbf{A}] = [XD(k, \ell)q] = a(k, \ell).$$

Similarly, define

$$\begin{aligned} \mathcal{B}_r &:= \sum_{i=1}^N y_i \ell_i r_i \prod_{j \neq i} k_j, \\ \mathcal{B}_s &:= \sum_{i=1}^N y_i \ell_i s_i \prod_{j \neq i} k_j. \end{aligned}$$

The relation $Y\ell = 0$ gives

$$u\mathcal{B}_r + v\mathcal{B}_s = 0,$$

and therefore

$$(3) \quad \mathbf{B} := \frac{\mathcal{B}_s}{u} = -\frac{\mathcal{B}_r}{v}$$

is a U -valued polynomial of bidegree

$$(N-2, 1).$$

It represents $b(k, \ell)$.

5.1. The base points. By (G1)–(G2), each coordinate form k_i has one zero $\kappa_i \in \mathbf{P}(K_X)$, and the κ_i are distinct. Similarly, ℓ_i has one zero $\lambda_i \in \mathbf{P}(K_Y)$, and the λ_i are distinct. Set

$$p_i := (\kappa_i, \lambda_i) \in \mathbf{P}(K_X) \times \mathbf{P}(K_Y).$$

Proposition 5.1 (Base locus). *For general X, Y , the base locus of each of the maps represented by $\mathbf{A}$ and $\mathbf{B}$ consists of the N reduced points*

$$p_1, \dots, p_N.$$

Blowing up these points resolves the pair map $\Phi_{X,Y}$.

Proof. We discuss **A**; the proof for **B** is symmetric.

First suppose all $\ell_i \neq 0$. If $\mathbf{A} = 0$, then

$$XD\ell' = 0$$

for a vector ℓ' independent from ℓ . Since

$$D\ell = k \in K_X,$$

we obtain $DK_Y \subseteq K_X$. If every $k_i \neq 0$, then D is invertible and $DK_Y = K_X$, contradicting (G3). If $k_i = 0$ for one index i , then DK_Y is contained in the coordinate hyperplane $\{t_i = 0\}$. Equality with K_X would contradict (G1). Thus there is no base point when all $\ell_i \neq 0$.

Now suppose $\ell_i = 0$. Since the other ℓ_j are nonzero, the cleared formula for **A** reduces, up to a nonzero scalar, to

$$x_i k_i.$$

It vanishes exactly when $k_i = 0$. Thus the set-theoretic base locus is $\{p_1, \dots, p_N\}$.

It remains to check that these base points are simple and that one blow-up resolves the map. Near p_i , use local coordinates

$$\xi = k_i, \quad \eta = \ell_i.$$

After multiplication by a local unit, the first-order part of the vector formula has the form

$$\xi c_i x_i + \eta Xc^{(i)},$$

where $c_i \neq 0$ and $c^{(i)} \in \mathbf{k}^N$ is supported away from the i -th coordinate.

The two vectors x_i and $Xc^{(i)}$ are independent for general (K_X, K_Y) . Indeed, if $Xc^{(i)} = \lambda x_i$, then

$$c^{(i)} - \lambda e_i \in K_X.$$

Using the ratios k_j/ℓ_j for $j \neq i$ and choosing the i -th diagonal entry so that a companion of ℓ maps to $c^{(i)} - \lambda e_i$, one obtains a diagonal matrix carrying K_Y to K_X . The entry is nonzero unless $\lambda = 0$. If $\lambda = 0$, then both $c^{(i)}$ and k belong to the one-dimensional space

$$K_X \cap \{t_i = 0\},$$

so $c^{(i)}$ is proportional to k . This would force K_Y to contain a coordinate vector, contradicting (G1). Hence $\lambda \neq 0$, and (G3) is contradicted.

Therefore the linear terms generate the maximal ideal (ξ, η) . Each base point is reduced, and on its exceptional divisor the two independent first-order vectors define a regular map. One blow-up resolves **A**, and symmetrically resolves **B**. $\square$

5.2. Pullback divisor classes. Let

$$\pi : S \longrightarrow \mathbf{P}^1 \times \mathbf{P}^1$$

be the blow-up of $p_1, \dots, p_N$. Let H_1, H_2 denote the pullbacks of the two ruling classes, and let $E_1, \dots, E_N$ be the exceptional curves. Thus

$$H_1^2 = H_2^2 = 0, \quad H_1 H_2 = 1, \quad E_i^2 = -1,$$

and all other pairwise intersections among these basis classes vanish.

Proposition 5.1 gives a morphism

$$\tilde{\Phi}_{X,Y} : S \longrightarrow \mathbf{P}(V) \times \mathbf{P}(U)$$

resolving $\Phi_{X,Y}$.

Proposition 5.2. *Let*

$$H_a := \tilde{\Phi}_{X,Y}^* \mathcal{O}_{\mathbf{P}(V)}(1), \quad H_b := \tilde{\Phi}_{X,Y}^* \mathcal{O}_{\mathbf{P}(U)}(1).$$

Then

$$H_a = H_1 + (N - 2)H_2 - \sum_{i=1}^N E_i,$$

and

$$H_b = (N - 2)H_1 + H_2 - \sum_{i=1}^N E_i.$$

Proof. The polynomial vector $\mathbf{A}$ has bidegree $(1, N - 2)$ and has the N simple base points p_i . Its proper-transform linear system therefore has the first displayed class. Formula (3) gives the second class. $\square$

6. MULTIDEGREE AND THE CHOW CLASS

Theorem 6.1 (Uniform frontier theorem). *Let $d \geq 2$, let $N = d + 3$, and let $X, Y \in (\mathbf{P}^d)^N$ be general. Then $\mathcal{C}_{d,N}(X, Y)$ is an irreducible rational surface. Its multidegrees are*

$$\deg_{2,0} = d - 1, \quad \deg_{1,1} = d^2 + d - 1, \quad \deg_{0,2} = d - 1.$$

Equivalently,

$$(4) \quad [\mathcal{C}_{d,d+3}] = (d - 1)h_a^d h_b^{d-2} + (d^2 + d - 1)h_a^{d-1} h_b^{d-1} + (d - 1)h_a^{d-2} h_b^d.$$

Its degree under the Segre embedding is

$$2d^2 + 4d - 4.$$

Proof. Theorem 4.2 proves irreducibility, rationality, and birationality of

$$\tilde{\Phi}_{X,Y} : S \longrightarrow \mathcal{C}_{d,d+3}(X, Y).$$

Consequently, the three multidegrees are the intersection numbers

$$H_a^2, \quad H_a H_b, \quad H_b^2$$

on S .

Using $N = d + 3$ and Proposition 5.2, we obtain

$$\begin{aligned} H_a^2 &= 2(N - 2) - N \\ &= N - 4 \\ &= d - 1, \end{aligned}$$

and similarly

$$H_b^2 = d - 1.$$

For the mixed intersection,

$$\begin{aligned} H_a H_b &= (N - 2)^2 + 1 - N \\ &= N^2 - 5N + 5 \\ &= d^2 + d - 1. \end{aligned}$$

These are the asserted multidegrees. Formula (4) follows by duality with the monomial basis of $A^*(\mathbf{P}^d \times \mathbf{P}^d)$.

Finally, the Segre hyperplane pulls back to $H_a + H_b$, so

$$\begin{aligned} \deg_{\text{Segre}} \mathcal{C}_{d,d+3} &= (H_a + H_b)^2 \\ &= H_a^2 + 2H_a H_b + H_b^2 \\ &= 2(d-1) + 2(d^2 + d - 1) \\ &= 2d^2 + 4d - 4. \end{aligned}$$

□

Remark 6.2 (Why the symmetry is now proved). The original plan expected the two outer coefficients of the Chow class to agree by symmetry. In Theorem 6.1, this is not an assumption: the two divisor classes H_a and H_b are exchanged by interchanging the two rulings and the two configurations, and their self-intersections are both $d-1$.

7. THE ONE-CENTER LOCI ARE RATIONAL NORMAL SCROLLS

Let

$$S_X := \overline{\pi_a(\mathcal{C}_{d,d+3}(X, Y))} \subset \mathbf{P}(V), \quad S_Y := \overline{\pi_b(\mathcal{C}_{d,d+3}(X, Y))} \subset \mathbf{P}(U).$$

Theorem 7.1. *For general X, Y , each of S_X and S_Y is a nondegenerate ruled surface of degree $d-1$ in $\mathbf{P}^d$. The maps*

$$\mathcal{C}_{d,d+3} \dashrightarrow S_X, \quad \mathcal{C}_{d,d+3} \dashrightarrow S_Y$$

are birational. Hence S_X and S_Y are rational normal scrolls, with the convention that a cone is allowed as a singular scroll.

Proof. The morphism $S \rightarrow S_X$ is given by H_a , and

$$H_a^2 = d-1 > 0.$$

Therefore its image is a surface and the map is generically finite. The image is nondegenerate: when $\ell_i = 0$ and $k_i \neq 0$, the formula for $\mathbf{A}$ reduces to x_i , so

$$X \subset S_X.$$

The general points $x_1, \dots, x_{d+3}$ span $\mathbf{P}^d$.

Let δ be the generic degree of $S \rightarrow S_X$. Then

$$\delta \deg S_X = H_a^2 = d-1.$$

Every nondegenerate irreducible surface in $\mathbf{P}^d$ has degree at least

$$\text{codim } S_X + 1 = d-1.$$

It follows that

$$\delta = 1, \quad \deg S_X = d-1.$$

For fixed $[\ell]$, formula (2) is linear in $[k] \in \mathbf{P}(K_X)$. Thus its image is generically a line. These lines rule S_X . The classification of varieties of minimal degree now identifies S_X as a rational normal scroll rather than the exceptional Veronese surface. The proof for S_Y is symmetric. □

Remark 7.2. The preceding theorem alone permits a cone. The next section computes the splitting type and rules out the cone for $d \geq 3$.

8. BALANCED SPLITTING AND ELEMENTARY TRANSFORMATIONS

Let

$$Q := \mathbf{P}(K_X) \times \mathbf{P}(K_Y) \cong \mathbf{P}^1 \times \mathbf{P}^1, \quad Z := \{p_1, \dots, p_N\} \subset Q,$$

and let

$$\rho_2 : Q \longrightarrow \mathbf{P}(K_Y)$$

be the second projection. Recall that the first center map is represented by the complete linear system of bidegree $(1, N-2)$ through Z . The word “complete” will follow in the proof below; it is important because it identifies the relevant ruled surface intrinsically.

Proposition 8.1 (The modification bundle). *For general X, Y , the sheaf*

$$\mathcal{E}_a := (\rho_2)_*(\mathcal{I}_Z \otimes \mathcal{O}_Q(1, N-2))$$

is a rank-two vector bundle on $\mathbf{P}(K_Y)$, and

$$\mathcal{E}_a \cong \mathcal{O}_{\mathbf{P}^1}(e) \oplus \mathcal{O}_{\mathbf{P}^1}(f), \quad e = \left\lfloor \frac{N-4}{2} \right\rfloor, \quad f = \left\lceil \frac{N-4}{2} \right\rceil.$$

In particular, $\mathcal{E}_a$ is balanced and

$$h^0(\mathcal{E}_a) = N-2 = d+1.$$

The symmetric bundle $\mathcal{E}_b$ has the same splitting.

Proof. Because the second coordinates $\lambda_1, \dots, \lambda_N$ are distinct, pushing forward

$$0 \longrightarrow \mathcal{I}_Z(1, N-2) \longrightarrow \mathcal{O}_Q(1, N-2) \longrightarrow \mathcal{O}_Z \longrightarrow 0$$

gives an exact sequence

$$(5) \quad 0 \longrightarrow \mathcal{E}_a \longrightarrow \mathcal{O}_{\mathbf{P}^1}(N-2)^{\oplus 2} \xrightarrow{\text{ev}} \bigoplus_{i=1}^N \mathbf{k}_{\lambda_i} \longrightarrow 0.$$

At λ_i , the quotient is evaluation at $\kappa_i \in \mathbf{P}(K_X)$. Thus $\mathcal{E}_a$ is the kernel of N negative elementary modifications. It is locally free of rank two and

$$\deg \mathcal{E}_a = 2(N-2) - N = N-4.$$

We now prove that the splitting is balanced rather than appeal to genericity informally. Twist (5) by $\mathcal{O}_{\mathbf{P}^1}(-t)$. For $0 \leq t \leq N-2$, the map on global sections is

$$\text{ev}_t : H^0(\mathcal{O}_{\mathbf{P}^1}(N-2-t))^{\oplus 2} \longrightarrow \mathbf{k}^N.$$

As K_X and K_Y vary, the ordered zero sets (κ_i) and (λ_i) vary dominantly in $(\mathbf{P}^1)^N \times (\mathbf{P}^1)^N$: one may prescribe the row linear forms of the two kernel matrices with these zeros. It is therefore enough to establish a nonempty open maximal-rank condition for general paired points. For general (κ_i, λ_i) , this map has maximal rank. To prove that the corresponding open rank condition is nonempty, specialize $\lceil N/2 \rceil$ of the quotient directions κ_i to $[1 : 0]$, the remaining $\lfloor N/2 \rfloor$ directions to $[0 : 1]$, and keep the λ_i distinct. Then ev_t is the direct sum of two ordinary polynomial-evaluation maps. If

$$s = N-1-t,$$

its rank is

$$\min\left(\left\lceil \frac{N}{2} \right\rceil, s\right) + \min\left(\left\lfloor \frac{N}{2} \right\rfloor, s\right) = \min(N, 2s).$$

This is maximal. Rank is an open condition. Since the parameter space is irreducible, this open set meets the open locus where all κ_i and all λ_i are distinct. Intersecting the finitely many open conditions for $0 \leq t \leq N-2$ proves simultaneous maximal rank.

It follows that

$$(6) \quad h^0(\mathcal{E}_a(-t)) = \max\{0, N - 2 - 2t\}.$$

Write

$$\mathcal{E}_a \cong \mathcal{O}_{\mathbf{P}^1}(e') \oplus \mathcal{O}_{\mathbf{P}^1}(f'), \quad e' \leq f'.$$

Together with $e' + f' = N - 4$, the values in (6) force

$$e' = \left\lfloor \frac{N-4}{2} \right\rfloor, \quad f' = \left\lceil \frac{N-4}{2} \right\rceil.$$

At $t = 0$, (6) gives

$$h^0(\mathcal{E}_a) = N - 2.$$

Finally, the $d + 1 = N - 2$ coordinate components of $\mathbf{A}$ are linearly independent: their image contains the general points $x_1, \dots, x_N$, which span $\mathbf{P}^d$. They therefore form a basis of

$$H^0(Q, \mathcal{I}_Z(1, N - 2)) = H^0(\mathbf{P}^1, \mathcal{E}_a).$$

Hence the first center map is indeed given by the complete linear system. The argument for $\mathcal{E}_b$ is symmetric. $\square$

Theorem 8.2 (Precise scroll type and contracted curves). *Let $d \geq 3$, $N = d + 3$, and let X, Y be general. Put*

$$e = \left\lfloor \frac{d-1}{2} \right\rfloor, \quad f = \left\lceil \frac{d-1}{2} \right\rceil.$$

Then

$$S_X \cong S(e, f) \subset \mathbf{P}^d, \quad S_Y \cong S(e, f) \subset \mathbf{P}^d.$$

Both scrolls are smooth.

On

$$S = \text{Bl}_{p_1, \dots, p_N}(Q),$$

define

$$C_i := H_2 - E_i, \quad D_i := H_1 - E_i.$$

The first center morphism contracts exactly the N disjoint (-1) -curves C_i , and the second contracts exactly the N disjoint (-1) -curves D_i . More precisely, there are factorizations

$$S \xrightarrow{\rho_a} S_X \hookrightarrow \mathbf{P}(V) \quad S \xrightarrow{\rho_b} S_Y \hookrightarrow \mathbf{P}(U),$$

where

$$\rho_a(C_i) = x_i, \quad \rho_b(D_i) = y_i.$$

Proof. We use the convention that $\mathbf{P}(\mathcal{E}_a)$ parametrizes one-dimensional quotients. The relative evaluation map associated to $\mathcal{E}_a$ produces the standard elementary transformation

$$S \longrightarrow \mathbf{P}(\mathcal{E}_a).$$

Locally over λ_i , all sections of $\mathcal{I}_Z(1, N - 2)$ vanish at κ_i . Consequently, the strict transform of the fiber

$$\mathbf{P}(K_X) \times \{\lambda_i\}$$

is contracted, while E_i becomes the new fiber. Thus the contracted curve is precisely

$$C_i = H_2 - E_i.$$

Away from these fibers, the relative degree-one system is an isomorphism onto a $\mathbf{P}^1$ -bundle. Hence no other curve is contracted by this elementary transformation.

Proposition 8.1 identifies the target as

$$\mathbf{P}(\mathcal{O}_{\mathbf{P}^1}(e) \oplus \mathcal{O}_{\mathbf{P}^1}(f)).$$

The tautological system maps it to the rational normal scroll $S(e, f)$. Since $d \geq 3$, one has $e \geq 1$, so the tautological system is very ample and the scroll is smooth. The component sections of $\mathbf{A}$ form the complete system by Proposition 8.1; hence this tautological map is exactly the first center map. At $\ell_i = 0$, the cleared formula for $\mathbf{A}$ is proportional to $x_i k_i$, so C_i is sent to x_i . The proof for the second center map and the curves D_i is symmetric. $\square$

9. CLOSED IMMERSION, SMOOTHNESS, AND THE GRAPH LINEAR SYSTEM

Theorem 9.1 (The resolved pair map is an embedding). *Under the hypotheses of Theorem 8.2, the morphism*

$$\tilde{\Phi}_{X,Y} : S \longrightarrow \mathbf{P}(V) \times \mathbf{P}(U)$$

is a closed immersion. Consequently,

$$\mathcal{C}_{d,d+3}(X, Y) \cong \text{Bl}_{p_1, \dots, p_{d+3}}(\mathbf{P}^1 \times \mathbf{P}^1)$$

is smooth, normal, and irreducible for every $d \geq 3$. Equivalently, it is the blow-up of either balanced scroll $S(e, f)$ at the $d+3$ points X , respectively Y .

Proof. By Theorem 8.2, the two coordinate maps are a blow-down of the C_i , respectively a blow-down of the D_i , followed by a closed scroll embedding.

We first separate points. A fiber of ρ_a is either a single point or one of the curves C_i . The restriction

$$\rho_b|_{C_i} : C_i \longrightarrow S_Y$$

is a closed immersion. Indeed,

$$C_i D_i = 0, \quad C_i D_j = 1 \quad (j \neq i),$$

and every intersection with D_j is transverse. Blowing down the disjoint D_j therefore maps C_i isomorphically to a smooth curve: locally, a curve transverse to an exceptional curve becomes a smooth curve through the blown-down point. Hence two points in one fiber of ρ_a are separated by ρ_b . The pair map is injective on geometric points.

We next separate tangent vectors. Away from the C_i , the differential of ρ_a is injective. Along a point of C_i not lying on a D_j , the morphism ρ_b is locally an isomorphism. At a point

$$C_i \cap D_j, \quad i \neq j,$$

the kernel of $d\rho_a$ is the tangent line to C_i , while the kernel of $d\rho_b$ is the tangent line to D_j . These two lines are transverse, so their intersection is zero. Finally $C_i \cap D_i$ is empty. Thus the pair differential is injective everywhere.

After composing with the Segre embedding, the resulting projective morphism separates geometric points and tangent vectors. Since S is proper, the projective embedding criterion implies that it is a closed immersion. Its image contains the dense open centers locus constructed in Theorem 4.2; hence its image is exactly the centers-variety. Smoothness, normality, and the two blow-up descriptions follow. $\square$

Corollary 9.2 (The birational map between the scrolls). *Let H be the hyperplane class and F a ruling fiber on S_X . The birational map*

$$S_X \dashrightarrow S_Y$$

whose graph is $\mathcal{C}_{d,d+3}(X, Y)$ is defined by

$$(7) \quad \left| (d+1)H + dF - d \sum_{i=1}^{d+3} x_i \right|.$$

The exceptional curve over x_i is mapped to a rational normal curve of degree d through

$$Y \setminus \{y_i\}.$$

The inverse statement is symmetric.

Proof. View S as the blow-up

$$\rho_a : S \longrightarrow S_X.$$

Its exceptional curves are $C_i = H_2 - E_i$. The pullback of H is H_a , and the pullback of F is H_2 . Since

$$\sum_{i=1}^N C_i = NH_2 - \sum_{i=1}^N E_i,$$

we obtain

$$H_1 = H_a + 2H_2 - \sum_{i=1}^N C_i.$$

Substitution into

$$H_b = (N-2)H_1 + H_2 - \sum_{i=1}^N E_i$$

gives

$$H_b = (N-2)H_a + (N-3)H_2 - (N-3) \sum_{i=1}^N C_i.$$

Using $N = d+3$, this is

$$H_b = (d+1)H_a + dH_2 - d \sum_{i=1}^N C_i,$$

which is the pullback of (7).

Moreover,

$$H_b C_i = d.$$

The image of $C_i \cong \mathbf{P}^1$ under the second scroll embedding is smooth by the proof of Theorem 9.1. It contains y_j for every $j \neq i$, because $C_i D_j = 1$ and D_j is contracted to y_j . These $d+2$ general points span $\mathbf{P}^d$. Hence the image is a nondegenerate smooth rational curve of degree d in $\mathbf{P}^d$, namely a rational normal curve. $\square$

Corollary 9.3 (Bigraded Hilbert polynomial). *For*

$$D = pH_a + qH_b,$$

one has

$$\begin{aligned} \chi(S, \mathcal{O}_S(D)) = 1 + \frac{1}{2} \Big(& (d-1)(p^2 + q^2) + 2(d^2 + d - 1)pq \\ & + (d+1)(p+q) \Big). \end{aligned}$$

In particular, the Hilbert polynomial for the Segre embedding is

$$\chi(S, \mathcal{O}_S(t(H_a + H_b))) = (d^2 + 2d - 2)t^2 + (d+1)t + 1.$$

Proof. The intersection numbers already computed give

$$D^2 = (d-1)(p^2 + q^2) + 2(d^2 + d - 1)pq.$$

The canonical class is

$$K_S = -2H_1 - 2H_2 + \sum_{i=1}^N E_i,$$

and direct intersection gives

$$DK_S = -(d+1)(p+q).$$

Since S is rational, $\chi(\mathcal{O}_S) = 1$. The surface Riemann–Roch formula

$$\chi(\mathcal{O}_S(D)) = \chi(\mathcal{O}_S) + \frac{1}{2}D(D - K_S)$$

proves both formulas. □

10. THE FIRST NEW CASE: $d = 4, N = 7$

We now specialize the general results to the case required by the research plan.

Theorem 10.1 (The $d = 4, N = 7$ centers surface). *Let X, Y be general ordered seven-point configurations in $\mathbf{P}^4$. Then $\mathcal{C}_{4,7}(X, Y) \subset \mathbf{P}^4 \times \mathbf{P}^4$ is a smooth irreducible rational surface, and the parametrization gives an isomorphism*

$$\mathcal{C}_{4,7}(X, Y) \cong \text{Bl}_{p_1, \dots, p_7}(\mathbf{P}^1 \times \mathbf{P}^1).$$

The two one-center loci are smooth cubic scrolls

$$S_X \cong S(1, 2), \quad S_Y \cong S(1, 2).$$

The centers surface is the blow-up of S_X at X , and also the blow-up of S_Y at Y . On the $\mathbf{P}^1 \times \mathbf{P}^1$ model, the two hyperplane classes are

$$H_a = H_1 + 5H_2 - \sum_{i=1}^7 E_i, \quad H_b = 5H_1 + H_2 - \sum_{i=1}^7 E_i.$$

Its Chow class is

$$[\mathcal{C}_{4,7}] = 3h_a^4 h_b^2 + 19h_a^3 h_b^3 + 3h_a^2 h_b^4.$$

Its multidegree is

$$(3, 19, 3),$$

and its degree in the Segre embedding

$$\mathbf{P}^4 \times \mathbf{P}^4 \hookrightarrow \mathbf{P}^{24}$$

is 44.

If H and F denote the hyperplane and ruling classes on S_X , then the birational map $S_X \dashrightarrow S_Y$ is defined by

$$\left| 5H + 4F - 4 \sum_{i=1}^7 x_i \right|.$$

Its exceptional curve over x_i maps to the rational normal quartic through $Y \setminus \{y_i\}$. The inverse statement is symmetric.

Finally, the bigraded Hilbert polynomial is

$$\chi(\mathcal{O}_{\mathcal{C}_{4,7}}(p, q)) = 1 + \frac{1}{2}(3p^2 + 38pq + 3q^2 + 5p + 5q),$$

and the Segre Hilbert polynomial is

$$22t^2 + 5t + 1.$$

Proof. Substitute $d = 4$ and $N = 7$ into Theorems 4.2, 6.1, 8.2, and 9.1, and into Corollaries 9.2 and 9.3. □

Remark 10.2 (Geometric interpretation). The expected surface does not fail: it exists and has only one irreducible component. Moreover, a general valid a on its cubic scroll determines a unique b on the other cubic scroll, and conversely. Thus the centers surface is the smooth graph closure of a birational correspondence between two cubic scrolls. Unlike the first version of this note, this is now a scheme-theoretic statement, not merely a birational description.

11. CONSISTENCY WITH THE KNOWN LOW-DIMENSIONAL CASES

The formula of Theorem 6.1 specializes as follows.

d	$N = d + 3$	multidegree	Segre degree	one-center locus
2	5	$(1, 5, 1)$	12	$S(0, 1) = \mathbf{P}^2$
3	6	$(2, 11, 2)$	26	$S(1, 1) \subset \mathbf{P}^3$
4	7	$(3, 19, 3)$	44	$S(1, 2) \subset \mathbf{P}^4$
5	8	$(4, 29, 4)$	66	$S(2, 2) \subset \mathbf{P}^5$

For $d = 2, N = 5$, Agarwal–Connelly–Crannell–Duff–Thomas prove that the centers surface is the graph of a degree-five Cremona transformation and has multidegree $(1, 5, 1)$ [1, Theorem 31]. For $d = 3, N = 6$, Ottaviani–Thomas prove that the centers surface is a smooth quadric blown up at six points; they compute multidegree $(2, 11, 2)$ and Segre degree 26 [5, Theorems 14 and 16]. Both agree exactly with the uniform kernel calculation above.

This agreement is a stringent check: the middle intersection number and the Segre degree depend on the base-point multiplicities, so they would change if the bihomogeneous degrees or the blow-up correction were incorrect.

There is a second, more sensitive check. When $d = 3$, write the quadric scroll as $\mathbf{P}^1 \times \mathbf{P}^1$ with

$$H = L_1 + L_2, \quad F = L_2.$$

Formula (7) becomes

$$4L_1 + 7L_2 - 3 \sum_{i=1}^6 x_i.$$

Up to interchanging the two rulings, this is exactly the graph linear system computed by Ottaviani–Thomas [5, Theorem 14]. Thus the new elementary-transformation calculation recovers not only their degrees but also the full divisor class of the birational map.

12. A PRIME IDEAL PRESENTATION AND THE REMAINING COMPUTATION

Write

$$\mathbf{A} = (A_0, \dots, A_d), \quad \mathbf{B} = (B_0, \dots, B_d)$$

for the component polynomials in (2)–(3). They belong to $\mathbf{k}[u, v, z, w]$ and have parameter bidegrees $(1, N - 2)$ and $(N - 2, 1)$.

Proposition 12.1 (Prime elimination presentation). *Let*

$$R := \mathbf{k}[a_0, \dots, a_d, b_0, \dots, b_d]$$

with target bigrading

$$\deg a_i = (1, 0), \quad \deg b_j = (0, 1).$$

Inside

$$\tilde{R} := R[u, v, z, w, s, t],$$

set

$$\mathfrak{J} := (a_i - sA_i : 0 \leq i \leq d) + (b_j - tB_j : 0 \leq j \leq d).$$

Then the bihomogeneous prime ideal of the centers-variety is

$$(8) \quad I(\mathcal{C}_{d+3}(X, Y)) = \mathfrak{J} \cap R.$$

Proof. The quotient $\tilde{R}/\mathfrak{J}$ is the polynomial ring $\mathbf{k}[u, v, z, w, s, t]$, under the identifications

$$a_i = sA_i, \quad b_j = tB_j.$$

It is a domain, so both $\mathfrak{J}$ and its contraction to R are prime. The affine variety parametrized by

$$(u, v, z, w, s, t) \longmapsto (s\mathbf{A}, t\mathbf{B})$$

is the affine cone over the image of the rational pair map. Values at the parameter base points contribute only the irrelevant cone vertex. The elimination theorem therefore identifies the contraction $\mathfrak{J} \cap R$ with the ideal of the projective image closure. By Theorems 4.2 and 9.1, that closure is the centers-variety. $\square$

Corollary 12.2 (The visible pure-factor equations in $d = 4$). *For $d = 4, N = 7$, after independent projective changes of the a - and b -coordinates, the ideal in (8) contains*

$$I_{S(1,2)}(a) + I_{S(1,2)}(b),$$

where each cubic-scroll ideal is generated by the three 2×2 minors of a matrix

$$\begin{pmatrix} x_0 & y_0 & y_1 \\ x_1 & y_1 & y_2 \end{pmatrix}.$$

Proof. The two coordinate projections of the centers surface are the cubic scrolls of Theorem 10.1. The displayed determinantal presentation is the standard ideal of $S(1, 2) \subset \mathbf{P}^4$. $\square$

12.1. What is solved and what remains. The previous scheme-theoretic questions are now resolved for $d \geq 3$: Theorem 8.2 classifies every curve contracted by an individual projection, and Theorem 9.1 proves point and tangent separation for the pair map. Hence there are no hidden boundary singularities: the centers-variety is smooth and already equal to the resolved surface.

Proposition 12.1 also gives an exact, saturation-free construction of the prime ideal. What remains is the minimal commutative algebra of this ideal. In the $d = 4, N = 7$ case one should now:

- (1) choose generic rational K_X, K_Y , form $\mathbf{A}, \mathbf{B}$, and execute the elimination in (8);
- (2) compute the minimal bihomogeneous generators and the bigraded Betti table;
- (3) compare the resulting Hilbert polynomial with

$$1 + \frac{1}{2}(3p^2 + 38pq + 3q^2 + 5p + 5q);$$

- (4) decide projective normality and the Cohen–Macaulay property of the chosen product embedding;
- (5) repeat over several large finite fields to separate stable generator patterns from unlucky specializations, and then prove the observed pattern in characteristic zero.

These are equation-minimization and syzygy questions. They are no longer needed to prove existence, irreducibility, smoothness, normality, scroll type, or multidegree.

12.2. Beyond the frontier. For $d \geq 4$,

$$D(d, d + 4) = 3 - d < 0.$$

Theorem 2.3 therefore gives

$$\mathcal{C}_{d,N}^\circ(X, Y) = \emptyset \quad \text{for general } X, Y \text{ and every } N \geq d + 4.$$

Thus $N = d + 3$ is the last generically nonempty case for $d \geq 4$. The exceptional low-dimensional cases $d = 2, 3$ have additional permitted values and are treated in [1, 5].

13. CONCLUSION

The research plan asked first for a rigorous dimension/emptiness framework and then for the first unknown case $d = 4, N = 7$. The universal incidence argument supplies the former. The two-dimensional kernels appearing precisely at $N = d + 3$ supply more: they give an explicit birational parametrization of every general frontier centers-variety.

The central new output is the uniform class

$$[\mathcal{C}_{d,d+3}] = (d - 1)h_a^d h_b^{d-2} + (d^2 + d - 1)h_a^{d-1} h_b^{d-1} + (d - 1)h_a^{d-2} h_b^d.$$

For $d = 4, N = 7$, this proves nonemptiness, purity, irreducibility, rationality, and the predicted class $(3, 19, 3)$. The continuation proves substantially more: for $d \geq 3$, the one-center surfaces are the balanced scrolls

$$S\left(\left\lfloor \frac{d-1}{2} \right\rfloor, \left\lceil \frac{d-1}{2} \right\rceil\right),$$

the resolved pair map is a closed immersion, and the centers-variety is smooth and normal. In the first new case it is the blow-up of the smooth cubic scroll $S(1, 2)$ at seven points, and the induced birational map of cubic scrolls is defined by

$$\left| 5H + 4F - 4 \sum_{i=1}^7 x_i \right|.$$

The prime ideal has the exact elimination presentation (8). The remaining work is now confined to finding minimal bihomogeneous generators, syzygies, and the bigraded Betti table.

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